Name: ______________________ Date: ________________

PROPORTION, (PROPORTION GRAPHS)

LO: I can interpret and draw graphs showing direct and inverse proportion.

Proportion graphs

A direct proportion graph is a straight line through the origin. An inverse proportion graph is a curve (hyperbola) that never touches the axes.

Key features: Direct: y = kx (linear, through origin). Inverse: y = k/x (curve, asymptotes at axes).

Quick examples

🟢	Direct proportion: straight line through (0, 0)	y = kx
🟢	Inverse proportion: curve approaching axes	y = k/x
🟢	Gradient of direct proportion line = k	constant rate
🔴	Any straight line shows direct proportion	must pass through origin

Key idea

Direct proportion: straight line through origin (gradient = k).
Inverse proportion: reciprocal curve.

Direct: $y = kx$ | Inverse: $y = \frac{k}{x}$

Recognise the shape to identify the type

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – direct proportion graph

The graph of y against x passes through (0, 0) and (4, 12). Is this direct proportion? Find k.

Passes through origin: yes, could be direct proportion

$k = y/x = \frac{12}{4} = 3$

$y = 3x$

Yes, k = 3

A straight line through the origin confirms direct proportion.

Example 2 – reading values from a graph

A direct proportion graph passes through (5, 20). Find y when x = 8.

$k = \frac{20}{5} = 4$

$y = 4x$

When x = 8: y = 32

y = 32

Use any point on the line to find k.

Example 3 – identifying inverse proportion

A graph curves downward, approaching but never touching the x-axis. The point (2, 6) is on it. Find the equation.

Shape suggests inverse proportion:

$y = k/x$

$k = xy = 2 \times 6 = 12$

$y = 12/x$

$y = \frac{12}{x}$

A curve with axes as asymptotes suggests inverse proportion.

Example 4 – comparing graph types

Graph A is a line through origin. Graph B is a curve approaching axes. Which shows direct proportion?

Direct proportion: straight line through (0, 0)

Graph A is the direct proportion graph

Graph B is inverse proportion

Graph A = direct, Graph B = inverse

The shape tells you the type of proportion.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying direct proportion

Does each graph show direct proportion? Explain.

a) Straight line through (0,0) and (3,9)

b) Straight line through (0,2) and (4,10)

c) Curve through (1,6), (2,3), (3,2)

d) Straight line through (0,0) and (5,15)

2. Finding equations from graphs

Find the equation of proportion from the given point.

a) Direct: passes through (4, 20)

b) Direct: passes through (3, 7.5)

c) Inverse: passes through (2, 10)

d) Inverse: passes through (5, 4)

3. Using proportion graphs

Use the given equation to find missing values.

a) $y = 4x$. Find y when $x = 7$

b) $y = 4x$. Find x when $y = 30$

c) $y = 24/x$. Find y when $x = 3$

d) $y = 24/x$. Find x when $y = 2$

4. Real-world proportion graphs

Identify the type and solve.

a) Cost vs quantity (straight line through origin). $k = 2.50$. Cost of 7 items?

b) Speed vs time for fixed distance. At 30 mph, takes 4 hours. Time at 50 mph?

c) Distance vs time at constant speed. Through (2, 100). Distance at $t = 5$?

d) Workers vs days (curve). 4 workers, 10 days. Days for 8 workers?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Direct proportion graphs pass through the origin ☐	**b) Any straight line graph shows direct proportion** ☐	**c) Inverse proportion graphs are curves that approach the axes** ☐	**d) Inverse proportion graphs cross both axes** ☐	**e) The gradient of $y = kx$ tells you k** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: _________________	Reason: _________________	Reason:	Reason: _________________	Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A taxi charges a **£3 fixed fee** plus **£1.50 per mile**. Draw the cost graph. Explain why this is NOT direct proportion despite being a straight line.

Expression: ____________________________

Simplified: ____________________________

b) Plot the points (1, 24), (2, 12), (3, 8), (4, 6), (6, 4), (8, 3). What type of proportion is this? Find the equation. Use it to predict: (i) y when $x = 12$, (ii) x when $y = 1.5$.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!